

Part 3: Software-Defined Vehicle (SDV) Centralization

Migrating from Distributed ECUs to High-Performance Computing (HPC)

1. Paradigm Shift: Legacy Distributed ECUs vs. SDV

Traditional vehicle electrical architectures are **distributed domain networks**. In this legacy model, each vehicular feature (door lock, wiper, seat motor, inverter, battery management) has its own dedicated micro-controller (ECU) supplied by different Tier-1 vendors with proprietary, static firmware.

A modern luxury vehicle often contains over **100 distributed ECUs**. This leads to massive architectural friction: implementing a feature requiring multiple domains (e.g. automatically locking doors and adjusting suspension when speed increases) requires coordinating software releases from multiple vendors.

Metric	Legacy Distributed Architecture	Software-Defined Vehicle (SDV)
ECU Count	70 - 120+ dedicated microcontrollers	1 - 3 Central Computers + Zonal Gateways
Control Logic	Burned into fixed ECU firmware	Consolidated in central software tasks
Wiring Harness	Heavy point-to-point domain wiring	Short zonal wiring to local edge hubs
Upgradability	Dealership flashing / physical recall	Over-The-Air (OTA) continuous updates
Feature Velocity	Years (hardware development cycle)	Weeks (agile software deployment)

2. Complete 10-Node Architecture Topology

Our SDV platform organizes all vehicle functions into two distinct virtual buses and 10 independent C processes:

Node Binary	ECU Subsystem	Bus Interface	Central Role & Responsibility
central_compute	Central Vehicle Computer (CVC)	vcan0 & vcan1	Dual-homed orchestrator; arbitrates torque, braking, steering, safety, thermal loops & displays live HUD.
pcm_node	Powertrain Control Module	vcan0	Simulates electric motor, inverter, vehicle inertia, aerodynamic drag, speed, and torque.
bms_node	Battery Management System	vcan0	400V traction pack, Coulomb-counting SoC %, pack current, and internal Joule heating.
brake_node	ABS & Stability Control	vcan0	Master cylinder pressure, 4-wheel slip detection, 15Hz ABS modulation pulsing, and rotor thermals.
eps_node	Electric Power Steering	vcan0	Speed-sensitive assist torque motor, driver hand torque sensor, and LKA overlay.
adas_node	Radar & Vision Perception	vcan0	Forward 77GHz radar, closing velocity, Time-To-Collision (TTC), FCW warnings, and AEB requests.
bcm_node	Body Control Module	vcan1	Door locks, low/high beams, brake lights, wipers, horn, and ambient light sensing.
hmi_node	Driver Cockpit Interface	vcan1	Simulates driver accelerator/brake pedals, steering wheel, gear selector, and drive modes.

hvac_node	Climate & Thermal Loop	vcan1	Dual cabin climate, evaporator, heat pump compressor load, and battery liquid coolant loop.
telematics_node	Telematics TCU & OTA	vcan1	Cellular 5G modem, GNSS RTK positioning, cloud telemetry ping, and A/B partition OTA updates.

3. Cross-Domain Central Software Orchestration

Crucially, **Bus 0 (`vcan0`)** and **Bus 1 (`vcan1`)** are never bridged directly. The Central Compute process acts as an active gateway and safety supervisor:

- **Autonomous Emergency Braking (AEB):** `adas\_node` on `vcan0` detects closing target and triggers AEB Level 2. Central Compute intercepts the signal, commands -180 Nm motor regen to `pcm\_node`, commands 2400 Nm hydraulic pressure to `brake\_node` (provoking ABS modulation), and flashes hazard lights via `bcm\_node` on `vcan1`.
- **Brake Blending:** Driver brake pedal (Bus 1) is dynamically arbitrated by Central Compute: light braking is harvested as electric regen (Bus 0), while heavy braking engages friction hydraulics (Bus 0).
- **Lane Keeping Assist (LKA):** Lane drift detected by ADAS vision (Bus 0) causes Central Compute to inject corrective torque overlay directly into `eps\_node` (Bus 0).
- **Battery Active Chilling:** When `bms\_node` (Bus 0) signals high cell temperatures, Central Compute commands `hvac\_node` (Bus 1) to engage chiller heat pump valves to cool the battery plate.
- **OTA Firmware Campaigns:** Cloud-directed firmware updates received by `telematics\_node` (Bus 1) are coordinated by Central Compute across all vehicle nodes.